INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

MODULE-1

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Project Name: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch -13

About the Project:

HotelRev Analyze is a data analytics project designed to track and evaluate hotel performance using booking and revenue data. The main goal is to study key performance indicators such as Occupancy Rate, Average Daily Rate (ADR), and Revenue Per Available Room (RevPAR) to understand how effectively hotel rooms are being used and how much income they generate.

In this project, raw booking data is first organized in Microsoft Excel, then cleaned and transformed using Power Query. After that, the data is further analyzed in Power BI by creating calculated measures and visual reports. This process turns unstructured data into clear, useful insights that highlight patterns in customer demand, pricing behavior, and seasonal trends.

The end result is an interactive Power BI dashboard that presents hotel performance through visuals like charts, KPI cards, and tables. These dashboards allow hotel management to easily track occupancy levels, compare revenue over different time periods, and make informed decisions to improve overall business performance.

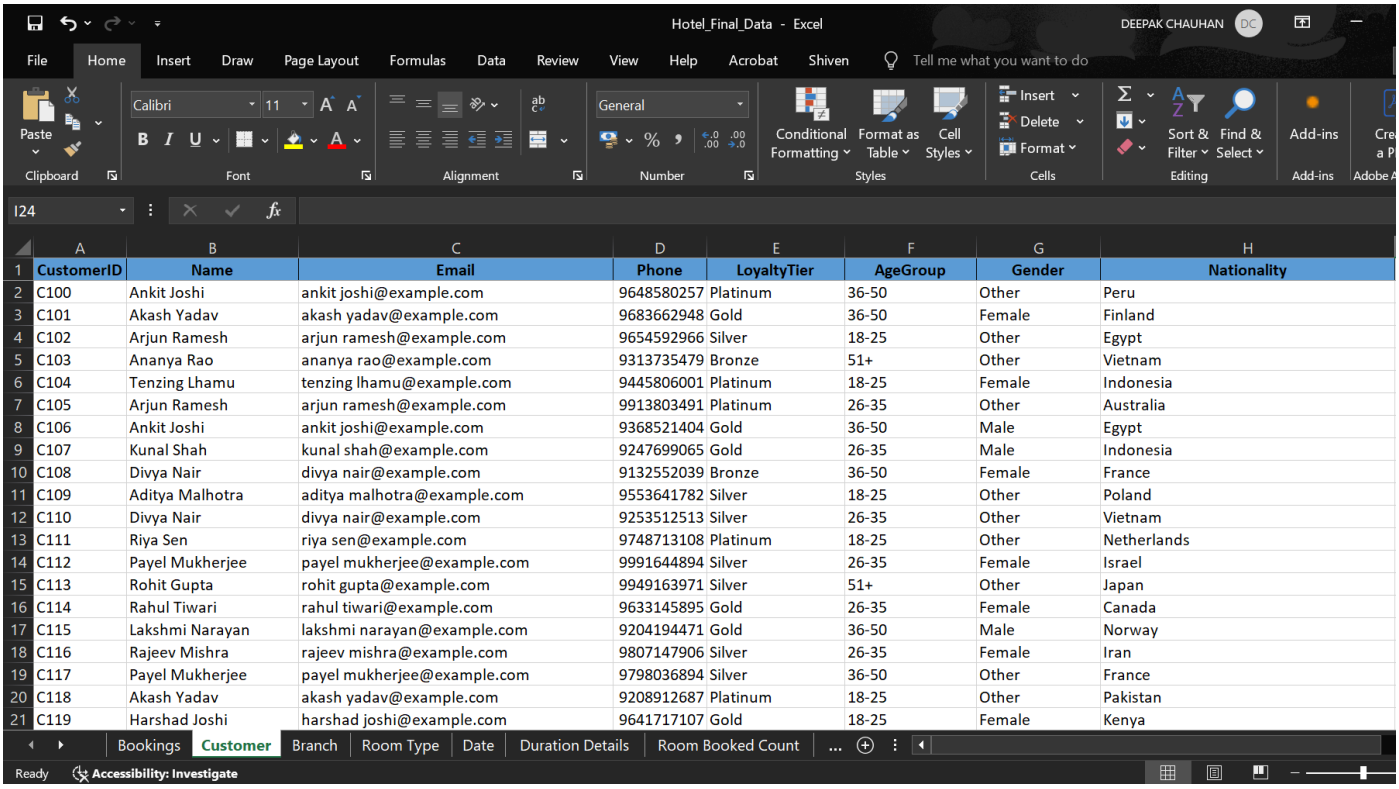
MODULE 1:INTRODUCTION AND DATA PREPARATION

Objective:-

Module 1 focuses on building a realistic hotel booking dataset with the help of Python. Instead of using existing data, this module creates synthetic records for different entities such as customers, hotel locations, room categories, and booking details, allowing the dataset to mimic real-world hotel operations.
Once the data is generated, it is structured and saved into CSV files for easy access and compatibility. These files act as the base layer for the next stages of the project, where the data is cleaned, transformed, and analyzed using tools like Power BI to extract meaningful insights.

Step-1: GENERATE BASIC DATASETS

The data for this project was created using a Python program designed to replicate real hotel booking activities. It produces random yet structured information for different components such as customers, hotel locations, room categories, and booking records. Libraries like csv, random, and datetime were used to generate and organize the data efficiently.
After generating the dataset, it was saved in CSV format, making it easy to import and work with in tools like Power BI. These files are then used in the next stage of the project for data cleaning, transformation, and detailed analysis.



CustomerID	Name	Email	Phone	LoyaltyTier	AgeGroup	Gender	Nationality
C100	Ankit Joshi	ankit joshi@example.com	9648580257	Platinum	36-50	Other	Peru
C101	Akash Yadav	akash yadav@example.com	9683662948	Gold	36-50	Female	Finland
C102	Arjun Ramesh	arjun ramesh@example.com	9654592966	Silver	18-25	Other	Egypt
C103	Ananya Rao	ananya rao@example.com	9313735479	Bronze	51+	Other	Vietnam
C104	Tenzing Lhamu	tenzing lhamu@example.com	9445806001	Platinum	18-25	Female	Indonesia
C105	Arjun Ramesh	arjun ramesh@example.com	9913803491	Platinum	26-35	Other	Australia
C106	Ankit Joshi	ankit joshi@example.com	9368521404	Gold	36-50	Male	Egypt
C107	Kunal Shah	kunal shah@example.com	9247699065	Gold	26-35	Male	Indonesia
C108	Divya Nair	divya nair@example.com	9132552039	Bronze	36-50	Female	France
C109	Aditya Malhotra	aditya malhotra@example.com	9553641782	Silver	18-25	Other	Poland
C110	Divya Nair	divya nair@example.com	9253512513	Silver	26-35	Other	Vietnam
C111	Riya Sen	riya sen@example.com	9748713108	Platinum	18-25	Other	Netherlands
C112	Payel Mukherjee	payel mukherjee@example.com	9991644894	Silver	26-35	Female	Israel
C113	Rohit Gupta	rohit gupta@example.com	9949163971	Silver	51+	Other	Japan
C114	Rahul Tiwari	rahul tiwari@example.com	9633145895	Gold	26-35	Female	Canada
C115	Lakshmi Narayan	lakshmi narayan@example.com	9204194471	Gold	36-50	Male	Norway
C116	Rajeev Mishra	rajeev mishra@example.com	9807147906	Silver	26-35	Female	Iran
C117	Payel Mukherjee	payel mukherjee@example.com	9798036894	Silver	36-50	Other	France
C118	Akash Yadav	akash yadav@example.com	9208912687	Platinum	18-25	Other	Pakistan
C119	Harshad Joshi	harshad joshi@example.com	9641717107	Gold	18-25	Female	Kenya

Customer.csv:

- CustomerID
- Name
- Email
- Phone
- Nationality
- Loyality tier
- Gender
- Age

Booking.csv:

- BookingID
- CustomerID
- RoomTypeID
- BranchID
- CheckInDate
- CheckOutDate
- Duration

- Revenue
- BookingStatus
- CancellationReason
- LeadTime
- PaymentMethod
- DiscountApplied
- BookingChannel

Branch.csv:

- BranchID
- BranchName
- City
- Phone
- ManagerName
- StarRating
- HasPool
- HasConferenceRoom

RoomType.csv:

- RoomTypeID
- RoomType
- Price
- MaxOccupancy
- Amenities
- IsAC
- IsSeaFacing

Step-2: Creating Additional coulmns required for Analysis.

In the Booking.csv file, additional fields such as *Stay Type* and *Room Nights Sold* were included to enhance the quality of analysis. These attributes help provide a clearer understanding of booking patterns and improve the accuracy of performance metrics.

	L	M	N	O	P	Q
1	PaymentMethod	DiscountApplied	BookingChannel	Purpose	StayType	RoomNightsSold
2	Cash	20	Travel Agent	Conference	Day Use	1
3	Cash	20	Mobile App	Business	Long Stay	7
5	Corporate Account	20	Mobile App	Holiday	Extended Stay	5
8	Credit Card	5	Website	Holiday	Extended Stay	4
9	UPI	20	Website	Business	Extended Stay	5
10	UPI	15	Website	Holiday	Extended Stay	4
11	Credit Card	20	Call Center	Conference	Long Stay	7
12	UPI	20	Call Center	Business	Long Stay	7
15	UPI	20	Mobile App	Conference	Short Stay	2
17	Cash	0	Website	Vacation	Extended Stay	4
19	Corporate Account	0	Travel Agent	Holiday	Long Stay	7
20	UPI	10	Website	Conference	Long Stay	7

StayType Formula:

=IF(H2=1,"Day Use",

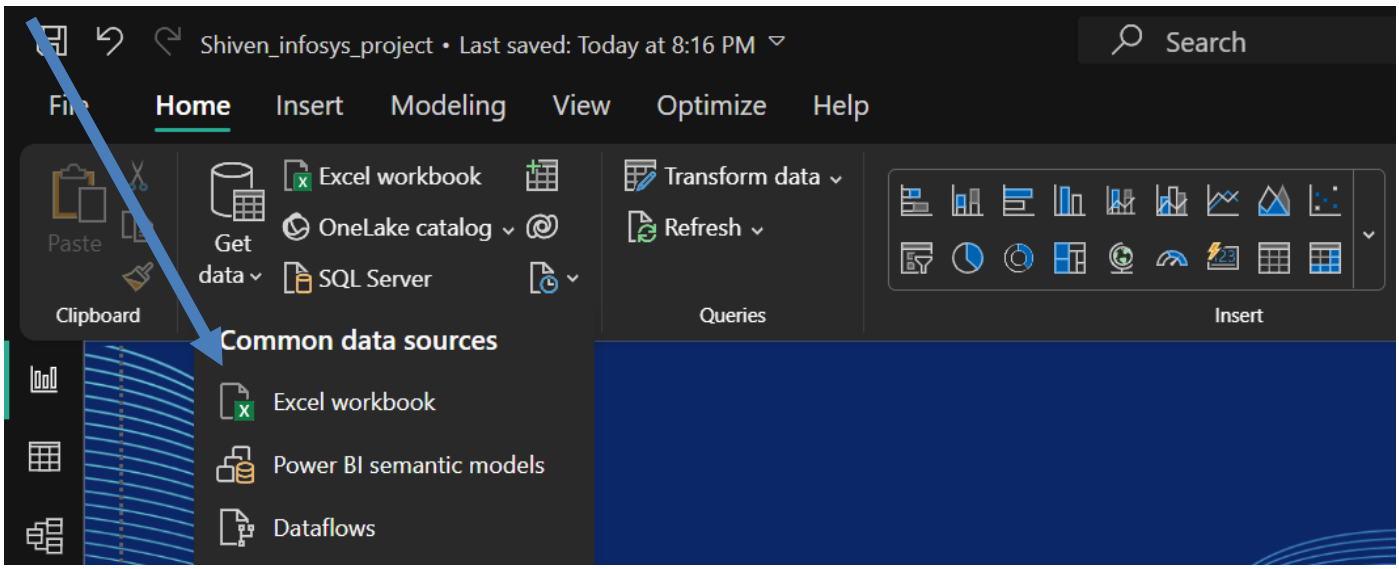
```
IF(H2<=2,"Short Stay",
  IF(H2<=5,"Extended Stay",
    "Long Stay")))
```

RoomNightSold Formula:

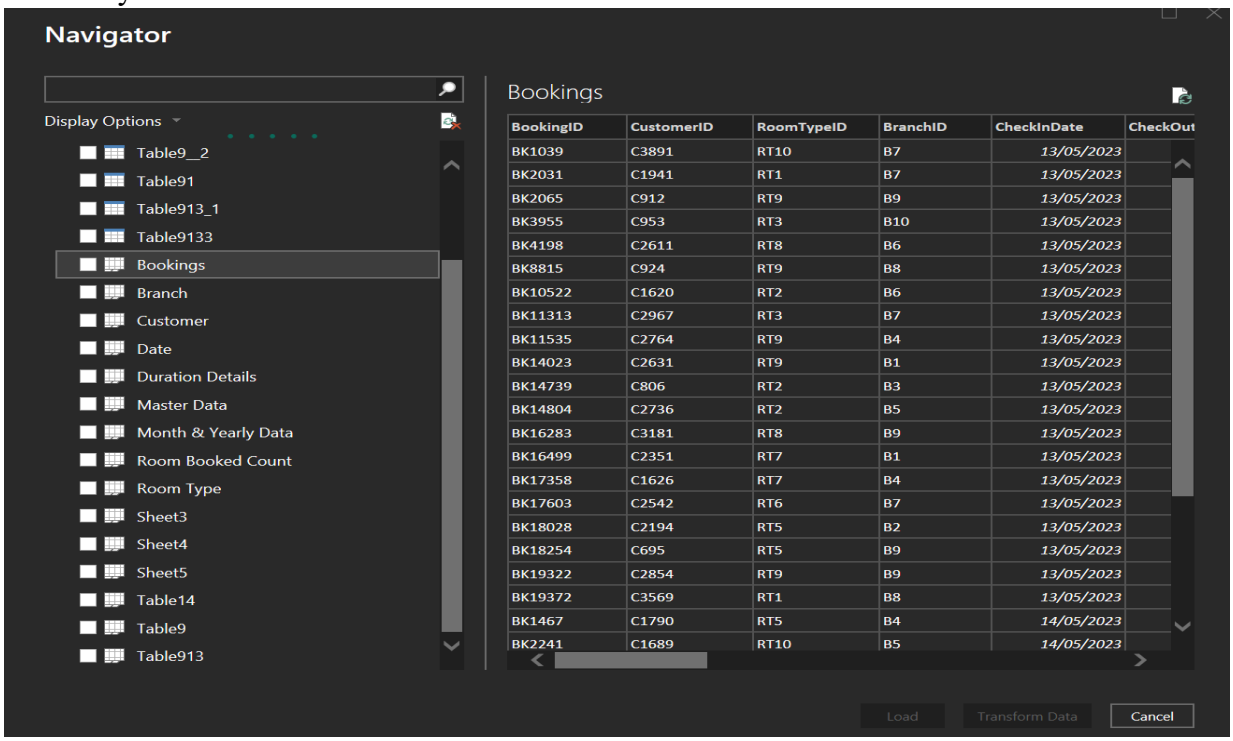
=SUMIFS(H2, J2, "Checked-In")

Step-3: Importing the data into PowerBI

The finalized dataset files were brought into Power BI using the **Get Data** feature. This step enables the data to be loaded into the platform, where it can then be cleaned, structured, and used to build models and visualizations for deeper analysis.



Select necessary tables and then click load

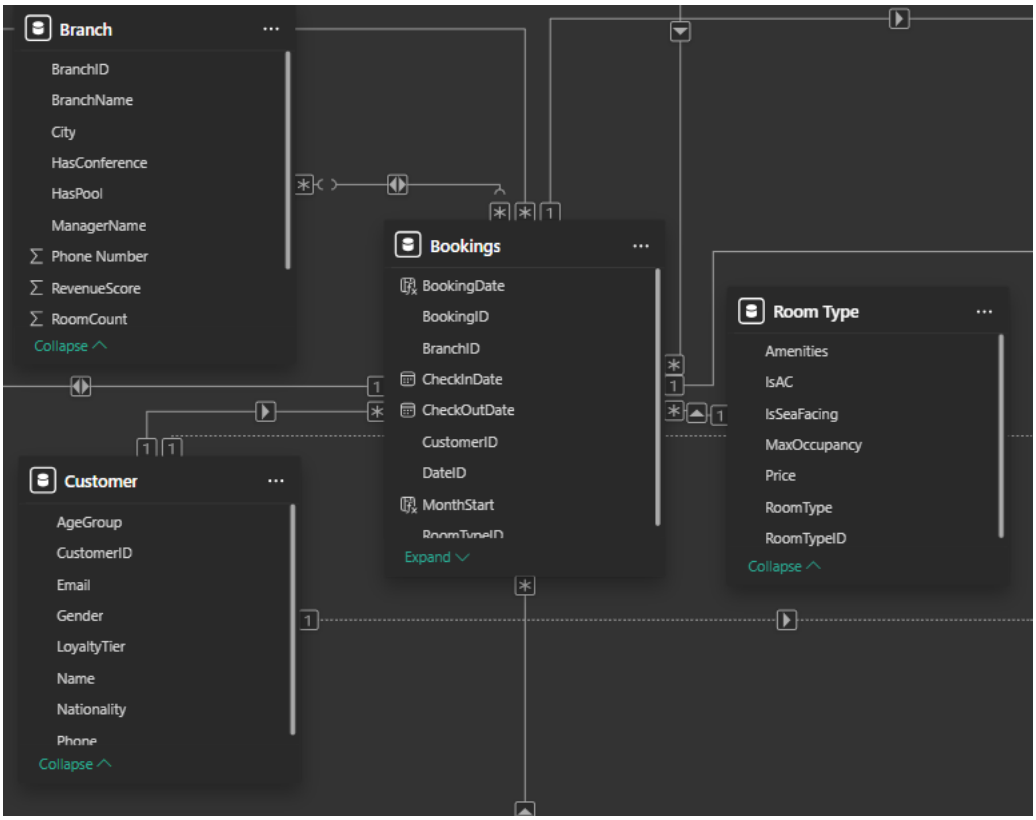


BookingID	CustomerID	RoomTypeID	BranchID	CheckInDate	CheckOut
BK1039	C3891	RT10	B7	13/05/2023	
BK2031	C1941	RT1	B7	13/05/2023	
BK2065	C912	RT9	B9	13/05/2023	
BK3955	C953	RT3	B10	13/05/2023	
BK4198	C2611	RT8	B6	13/05/2023	
BK8815	C924	RT9	B8	13/05/2023	
BK10522	C1620	RT2	B6	13/05/2023	
BK11313	C2967	RT3	B7	13/05/2023	
BK11535	C2764	RT9	B4	13/05/2023	
BK14023	C2631	RT9	B1	13/05/2023	
BK14739	C806	RT2	B3	13/05/2023	
BK14804	C2736	RT2	B5	13/05/2023	
BK16283	C3181	RT8	B9	13/05/2023	
BK16499	C2351	RT7	B1	13/05/2023	
BK17358	C1626	RT7	B4	13/05/2023	
BK17603	C2542	RT6	B7	13/05/2023	
BK18028	C2194	RT5	B2	13/05/2023	
BK18254	C695	RT5	B9	13/05/2023	
BK19322	C2854	RT9	B9	13/05/2023	
BK19372	C3569	RT1	B8	13/05/2023	
BK1467	C1790	RT5	B4	14/05/2023	
BK2241	C1689	RT10	B5	14/05/2023	

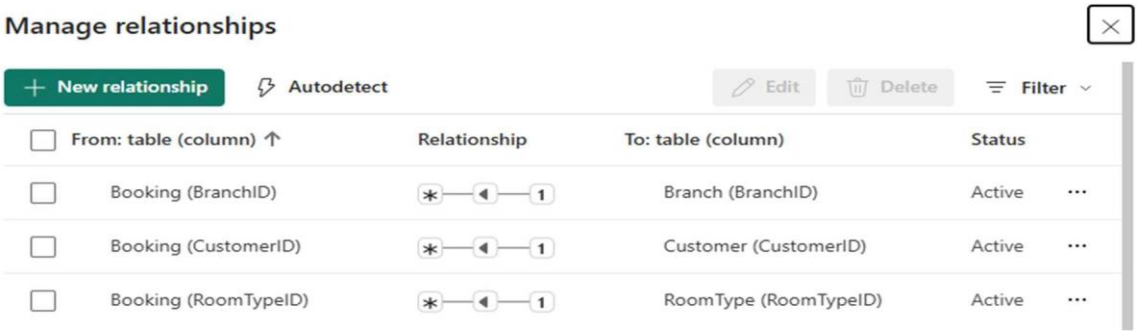
Step-3: Create STAR schema in PowerBI

Once the datasets were loaded into Power BI, a star schema model was designed to structure the data effectively. The central fact table (Booking) was linked to supporting dimension tables like **Customer, Branch**, and **Room Type**.

This modeling approach keeps the data well-organized and reduces complexity, making it easier to analyze. It also improves report performance and ensures faster, more efficient insights during visualization and reporting.



Relationships between the tables as given in the following figure:



Conclusion:

In this module, all the necessary datasets for the hotel revenue analysis project were successfully created using Python. To enhance the data for analysis, additional columns were added with the help of Excel formulas. After preparing the datasets, they were imported into Power BI for further processing.
A star schema data model was then built to structure the data in a clear and efficient way. This setup makes it easier to perform transformations, carry out analysis, and create meaningful visualizations in the later stages of the project.